NARRA-SCALE: Scaling Users and Messaging through Narrative Detection in Retweet Networks

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***Abstract*—In politically charged environments, understanding how ideological narratives emerge, spread, and shape user behavior on social media is critical for applications ranging from misinformation detection to enriching public discourse with verifiable truth. In this study, we present NARRA-SCALE, a framework that brings together network analysis, narrative detection, stance classification, and bipartite scaling to place users, communities, and messages along a single ideological dimension. As a case study, we apply NARRA-SCALE to a U.S. race relations related dataset chiefly polarized between "Black Lives Matter" and "All Lives Matter" supporters. We extract topic coded key phrases and named entities using frequency-based heuristics. Using key phrase co-occurrence relationships and latent representations of matching messages, we mine grouped (entities, issues/aspects, values) triplets characterizing key recurring narratives within the corpus. We use an LLM to summarize messages matching each triplet. Subsequently, a panel of experts label the stance information of the narratives on their key phrases, enabling weak supervision for training a high-accuracy stance detection model which achieves an 81% F1 score on a held-out gold standard. Next, we construct a signed bipartite graph with colored edges (i.e., representing support versus opposition) between users and key terms mentioned in their messaging corresponding to debated core values, issues, and actors to co-scale their positions on a [-1,+1] range. Our method reaches 91% agreement with user groups identified through community structure as well as with the "ideal points" of political elites and the general public on Twitter in the U.S. and five European countries.**

***Index Terms*—social networks, political polarization, narrative extraction, stance detection, ideological scaling**

I. Introduction

Social media has reshaped the way ideological narratives take shape, spread, and influence public debate, especially on politically sensitive issues. In polarized contexts, platforms such as Twitter (now X) act both as arenas where competing narratives gain visibility and as valuable sources of data for computational research. Uncovering the underlying structure of ideological alignments and tracing how narratives evolve at scale remains difficult because narratives shift quickly, explicit stance cues are sparse, and the data itself is noisy.

Research on polarization in online platforms has become central to understanding contemporary public discourse. Twitter, in particular, has been shown to host strongly divided communities, where retweets, follower ties, and interaction patterns often mirror clear ideological boundaries. [1] demonstrates that political leanings can be inferred from these networks, with users tending to engage mostly with like-minded others, reinforcing "echo chambers."

On social media, narratives are not just single posts or hashtags. They form connected stories that shape how people understand political events and social issues. Furthermore, while hashtags may signal affiliation, they do not necessarily encapsulate the broader semantic and emotional context of a narrative. To overcome these limitations, models need to account for both the language used and the network structure of political discourse.

In highly controversial topics such as racial justice or election integrity, opposing ideological camps promote divergent narratives that compete for visibility and influence. For example, the "Black Lives Matter" (BLM) and "All Lives Matter" (ALM) movements illustrate antagonistic narrative frameworks that not only differ in messaging but also reflect deeper schisms [2]. Previous work [3] has shown that viral messages can function as rallying points for collective identity formation and political mobilization.

In this paper, we introduce *NARRA-SCALE*, a scalable framework that combines: (i) network analysis to detect communities in retweet graphs, (ii) key phrase and named entity extraction, (iii) narrative modeling with Boolean triplet patterns, (iv) weakly supervised stance classification, and (v) bipartite partitioning and ranking to co-scale users and messaging elements on a continuous ideological axis. We apply *NARRA-SCALE* to a polarized U.S. race-relations dataset comprising over 14 million tweets from May 2023.

This study makes three core contributions: • Interpretability: We introduce a novel approach to patterned narrative detection and modeling, leveraging interpretable Boolean triplets that integrate key actors, topic-coded issues, and fundamental human values. • Scalability: NARRA-SCALE achieves effective socio-cultural and political scaling across vast social media networks, providing robust analysis at both the individual user and narrative levels. • High Accuracy: The framework demonstrates a 91% agreement, robustly correlating with both user camps inferred from community structures and the "ideal points" extrapolated from a substantial sample of political elite and mass public Twitter users [1] across the United States and five European nations.

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II. Related Works

A. Community Detection and Polarization in Social Networks

Community detection in Twitter retweet networks has been central to understanding political polarization. [4] is among the first to reveal that Twitter networks split into ideologically homogeneous clusters using modularity-based algorithms. Subsequent work by [5] further demonstrate that the Louvain algorithm reliably extracts communities in retweet networks that align with real-world political groups, confirming the structural echo chambers in online discussions. Studies [6], [7] advance this line of research by exploring retweet overlap networks and proposing graph-based controversy scores to quantify polarization beyond simple community detection.

Prior work on community detection in social networks has extensively leveraged algorithms such as Louvain [8] to uncover cohesive user communities, particularly within retweet networks where polarization and ideological clustering are prominent. Studies like [9]–[13] apply modularity-based and label propagation methods to reveal the partisan structure of Twitter interactions.

More recent studies have improved community detection by taking both sentiment and content into account. [14] use sentiment-weighted mention networks during Ireland's 2018 abortion referendum to reveal sharply polarized groups, while [15] show that Norwegian Twitter remains cross-cut despite ideological clustering. Extending these insights, [16] present Retweet-BERT, which blends linguistic homophily with retweet structures to uncover echo chambers in COVID-19 debates, and [17] introduce HyperGraphDis, a hypergraph model integrating cascades, user ties, and semantics for scalable, state-of-the-art disinformation detection.

B. Narrative, Framing, and Discourse Analysis

Narrative extraction diverges fundamentally from topic modeling: whereas topic modeling uncovers the dominant themes within a corpus [18], narrative extraction seeks to assemble coherent story elements—events, actors, relationships and their temporal order—to reconstruct an underlying "storyline" or plot [19]. Beyond purely graph-based approaches, recent work has probed the micro-level framing and discourse properties of social media posts. Frame-classification studies demonstrate that even within 280 characters, distinct rhetorical frames (e.g., moral versus economic emphasis) can be detected by joint models that learn latent frame embeddings alongside lexical indicators [20]. Such analyses reveal *how* narratives are linguistically packaged to persuade or mobilize different audience segments.

Hybrid methods combine topic modeling with large language models to refine narrative detection. Narrative Shift Detection uses Dynamic Topic Models to capture evolving thematic distributions over time, then applies large language models to semantically label and refine those shifts into coherent narrative events [21]. More recently, empirical work on narrative perception has shown that lay annotators' judgments about whether a social media post constitutes a "story" vary according to features like plot complexity and cohesion, suggesting that future narrative extraction models must account for reader-centric variability as well as computational criteria [22]. In parallel, [23] introduce a few-shot, domain-agnostic approach that uses Natural Language Inference to evaluate whether text supports or contradicts predefined claims, reducing annotation costs while maintaining strong performance across tasks.

Other approaches leverage community and graph signals to surface story elements. [24] propose a narrative-map optimization that extracts "accepted" social movement narratives across Reddit communities during the 2021 Cuban protests, revealing how different subreddits emphasize distinct events and grievances. At a broader scale, [25] construct temporal knowledge graphs across platforms (Instagram, TikTok, Twitter, YouTube) to mine recurring cross-platform narrative templates in Asia–Pacific political discussions, identifying coordinated sequences of events used to amplify competing viewpoints. Building on this line of work, [26] examine UK parliamentary debates using LLM-based stance labeling and a semi-automated method for extracting narrative frames, showing how migration discourse has gradually shifted toward security concerns and rights-based framings.

Alongside these framing methods, various pipelines have been built to detect narratives as they emerge on social media. TrollHunter2020, for example, continuously monitors election-related tweets via Twitter's streaming API and flags emerging "trolling" storylines through sliding-window term co-occurrence and semantic-embedding clustering [27]. Similarly, Happenstance employs an embedding-based semantic-search pipeline to trace Russian state-media narratives as they infiltrate Reddit discussions, quantifying how propaganda topics propagate into specific online communities [28]. In a related direction, [29] present TwiXplorer, an open-source dashboard that combines narrative detection with topic and network analysis of large-scale historical Twitter datasets.

C. Bipartite Co-Scaling and Ideological Inference

Bipartite co-scaling methods seek to jointly embed users and political entities (e.g., legislators, hashtags, tweets) into a common ideological space by decomposing a two-mode user–entity matrix. In one of the foundational studies, [1] models the Twitter followership network between ordinary users and elected officials as a bipartite graph and applied a Bayesian ideal-point estimation to infer each node's position on a latent liberal–conservative dimension. The resulting user and legislator scores closely replicate roll-call based measures, demonstrating that follow links alone suffice to recover meaningful ideological orderings. The ideal-point scores (θ) computed via Barberá's Bayesian estimation have been directly utilized in multiple studies [30]–[35]. Building on this idea, [36] construct a dynamic lexicon of partisan-charged terms and assemble a user–term contingency table; correspondence analysis is then used to co-scale citizens, elites, and lexicon items along a unified ideological axis, yielding validated estimates of individual-level ideology from text alone. Similarly,

[37] address the challenge of scaling ideological estimates to the full Twitter population. Starting from a seed set of users with known elite-derived positions, they apply homophily- and diffusion-based propagation over the user–elite bipartite graph to infer ideology for millions of additional accounts.

More recent work has extended bipartite co-scaling via graph-based neural architectures. [38] propose a framework, which embeds users and political actors within a multi-relational bipartite graph—incorporating follow, retweet, mention, and like edges—using a graph-neural network. The framework jointly optimizes across all relations to learn low-dimensional user embeddings that improve ideology classification beyond monolithic co-scaling approaches. Similarly, [39] introduce a bipartite GNN that separately aggregates homogeneous (user–user) and heterogeneous (user–tweet) neighborhoods before fusing them to predict users' Democratic vs. Republican leanings, on data from the 2020 U.S. election.

In parallel to correspondence-analysis approaches, link-analysis ranking algorithms have been adapted to bipartite political graphs to derive continuous ideology scores for users. Classic PageRank [40] and HITS [41] assign centrality values by iteratively redistributing scores over network links; their bipartite extensions, Co-HITS [42] and ANCO-HITS [43], alternate score propagation between user nodes and political entity nodes (e.g., legislators, hashtags, tweets) while applying normalization to account for polarization or signed edges. When applied to a user–politician follow graph on Twitter, Co-HITS produces dual hub/authority scores that co-locate both users and elites in a shared ideological space, and ANCO-HITS further refines this by enforcing a single continuous dimension that aligns with the liberal–conservative spectrum.Building on elite-anchored scaling, personalized PageRank and label-propagation techniques have been used to diffuse a small set of known ideology scores through the follower network, effectively scaling ideology to the entire user population [44].

III. Data & Methodology

We analyze a large-scale Twitter dataset from May 2023 on U.S. race relations, consisting of over 14 million tweets produced by approximately 3.7 million unique users. The collection spans original posts, retweets, replies, and quote tweets. Among these, we identify $\sim$5.9M distinct content tweets generated by $\sim$3.7M unique users.

NARRA-SCALE framework integrates network-based community detection, named entity and key phrase extraction, aspect-based sentiment analysis [45], to systematically detect and classify user stances as pro- or anti- towards detected terms comprising named entities and topic-coded issues, aspects, and values. Next, using a partitioning and scaling algorithm [43], we rank the network objects by the magnitude of their political polarity on a $[-1, +1]$ range.

A. Retweet Network Construction and Community Detection

We construct a retweet-based interaction network in which users are nodes and retweet relationships form weighted edges. To uncover ideological structure, we apply the Louvain algorithm [8], which partitions the network into cohesive clusters with high modularity. We identify seven major communities and further refine them into 25 subcommunities (Figure 1).

Following that, a panel of domain experts reads and analyzes the top 10 most viral tweets from each subcommunity, mapping each to one of the underlying ideological camps (i.e., ALM or BLM) (Figure 1 (b)). This subset of the dataset is used as the labeled corpus for further analysis.

B. Key Phrase and Named Entity Extraction

To extract meaningful terms from the dataset, we use a set of keyword extraction tools—spaCy [46], Textacy [47], RAKE [48], and YAKE [49]—to identify key noun phrases and named entities. Noun phrases are uniquely central to discourse, being the sole elements unambiguously classifiable into main topics [50]. They provide more salient information than verbs, generally governing verb phrase expression [51]. Following noun phrase extraction, we also apply a filtering process to remove dates, numbers, stop words, and other generic ambiguous phrases.

Because our dataset includes both single words and multi-word phrases, common unigrams like "immigration" may appear more frequent than more informative phrases such as "illegal immigration," even when the latter occurs more often than other uses of "immigration." To address this, we apply discounting that adjusts unigram frequencies by subtracting the counts of overlapping longer n-grams whenever their frequencies exceed the discounted value of the unigram. We then use the elbow method to decide a cutoff point, selecting the most informative set of phrases and entities to include in the narrative detection analysis.

C. Patterned Narrative Detection and Stance Labeling

Social media feeds present a constant stream of competing content. Recent studies [52], [53] show that moral and emotional messages capture attention more quickly than neutral ones and are more likely to spread widely during online political discussions.

Our approach integrates rule-based lexical classification, network-analytic centrality measures, and unsupervised clustering in an end-to-end pipeline to detect moral and emotional narratives. We commence by constructing a weighted, undirected graph, denoted as $G = (V, E)$, where each node $v \in V$ represents a lexeme of interest encompassing named entities, salient issues or aspects, and phrases intrinsically linked to core human values such as Christianity, Democracy, Environment, Family, Homeland, Justice, Life, and Truth, as identified within our corpus. Each edge $(u, v) \in E$ is weighted by the frequency with which the lexemes u and v co-occur within the same tweet. Formally, if f_{uv} denotes the number of tweets containing both u and v, then the graph weight is defined as

$$w(u, v) = f_{uv} \,.$$

Edges with $w(u, v) < \tau$, where τ is the parametric threshold, are pruned to reduce noise.

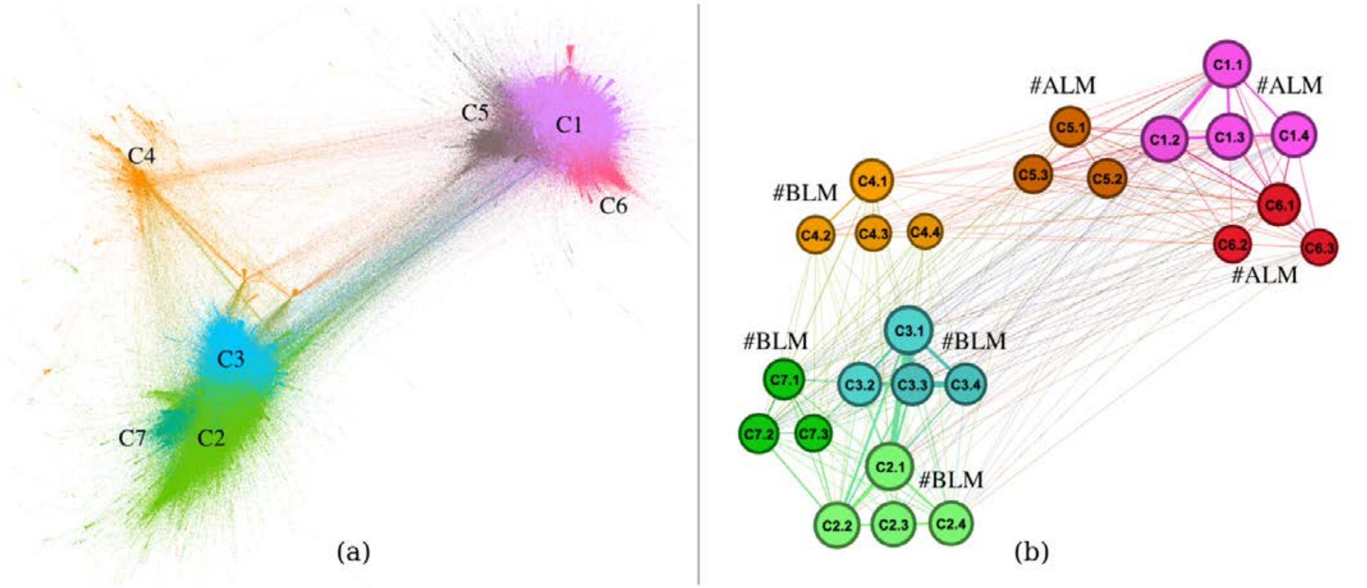

Fig. 1. Retweet network communities detected using the Louvain algorithm. (a) Visualization of seven top-level communities (C1–C7), each represented by a distinct color. (b) Refinement into 25 subcommunities, with expert labels indicating alignment to the dominant ideological camps (#ALM or #BLM).

To quantify the structural importance of each edge, we compute the edge betweenness centrality (EBC) $\beta(e)$ for each $e \in E$ in G [54]. Edge betweenness centrality is defined as

$$\beta(e) = \sum_{s \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}},$$

where σ_{st} is the total number of shortest paths between nodes s and t, and $\sigma_{st}(e)$ is the number of those paths that traverse edge e. Higher $\beta(e)$ values indicate edges that act as critical "bridges" connecting different dense subgraphs, and are thus more salient for narrative linkage.

We focus on moral and emotional triplet motifs of the form $(\mathrm{NE} \leftrightarrow \mathrm{IS/AS} \leftrightarrow \mathrm{VAL})$, where:

- NE denotes a named entity (e.g., "america", "republicans"),
- IS/AS denotes an issue or an aspect (e.g., "racism", "police brutality"),
- VAL denotes a core human value term (e.g., "equality", "christians").

We enumerate all concatenations of two adjacent edges $\big(\mathrm{NE}, \mathrm{IS/AS}\big)$ and $\big(\mathrm{IS/AS}, \mathrm{VAL}\big)$. Each candidate triplet $T = (\mathrm{NE}, \mathrm{IS/AS}, \mathrm{VAL})$ is assigned a composite score

$$S(T) = 2^{\beta(\mathrm{NE},\mathrm{IS/AS}) + \beta(\mathrm{IS/AS},\mathrm{VAL})},$$

which amplifies the contribution of highly central edges. We retain only those triplets for which all three terms co-occur in at least one tweet and whose aggregate co-occurrence frequency exceeds a minimum support threshold.

To capture the semantic context of each triplet, we employ a pre-trained transformer encoder, BERT, to generate contextual embeddings. For each tweet d containing a given triplet T, we compute an embedding vector $\mathbf{e}_d$ by averaging its token-level representations. We then obtain a triplet-level embedding $\mathbf{e}_T$ as the mean of the embeddings for tweets $d \ni T$:

$$\mathbf{e}_T = \frac{1}{|D_T|} \sum_{d \in D_T} \mathbf{e}_d,$$

where D_T is the set of tweets in which all three terms of T co-occur. We apply UMAP to reduce $\{\mathbf{e}_T\}$ for clustering. Finally, we cluster the UMAP-projected triplet embeddings using DBSCAN [55]. The DBSCAN algorithm groups triplets into dense regions of semantic similarity, discovering coherent "narrative clusters" without prespecified labels. The resulting clusters $C_1, C_2, \ldots, C_k$ represent distinct narrative threads characterized by their constituent triplets and central lexical motifs.

Next, we provide the LLM with each narrative cluster, along with triplets and matching sample tweets, to generate human-readable summaries of the narratives. The exported narratives along with supporting tweets enable domain experts to qualitatively interpret the results. The words within the triplet combinations corresponding to each narrative are then labeled as supported vs. opposed, or left blank by domain specialists, based on their review of the generated summaries and matching messages, to curate a weakly labeled dataset for the next steps. Table I demonstrates the most viral pair of narratives belonging to the ALM and BLM camps with anonymized tweets matching the narrative pattern.

D. Weak Supervision and Model Training for Stance Detection

Our patterned narrative detection algorithm extracts 52 distinct narratives-20 from the ALM camp and 32 from the BLM camp. The matching dataset, encompassing these 52 narratives, comprises 191,643 tweets from the ALM camp and 449,777 from the BLM camp. Subsequently, our team of

ALM.N.1.White Supremacy as a Weaponized Narrative in Modern America
Narrative summary: The media and political elites are using the white supremacy label to distract from border crises and rising crime, even falsely applying it to non-white individuals to fuel division and control the narrative.
Pattern: ['white supremacist', 'white supremacy']↔['racist', 'blacks', 'black people', 'woke', 'white people', 'whites', 'citizen', 'violence', 'latino', 'work', 'white person', 'white privilege', 'culture', 'media']↔['lie', 'fact', 'child', 'kids', 'son', 'election', 'love', 'border', 'god', 'free speech', 'patriot', 'christian', 'family', 'laws', 'nation', 'truth', 'hell', 'vote', 'parent', 'life']


1,000,000 people a year invading and replacing the home population (of all skin colours) who are being called racists and white supremacists for objecting.
No border controls means no functioning welfare state.
Can't have both.
Protect the country which is your home.
5:21 PM · May 21, 2023
562 Retweets 562 Quote Tweets 2.6K Likes


They're calling the Texas shooter a "white supremacist" because they don't want Americans to know about violent drug cartels running human trafficking operations on our border
4:07 PM · May 23, 2023
4.1K Retweets 471 Quote Tweets 20K Likes


So the media is stating that the Allen Texas Mall shooter (Mauricio Garcia) was a right wing, white supremacist.
Really? The guy didn't have a job, his parents couldn't speak English, he was paying for a hotel room, drove a Dodge Charger, and could afford tactical gear plus an AR-15.
Thoughts?
10:44 PM · May 7, 2023
2.9K Retweets 1.6K Quote Tweets 7.7K Likes

ALM.N.2.Racial Violence and Political Hypocrisy in America
Narrative summary: Democrats have used systemic racism and Black voters for political gain, while white Americans are now the primary victims of racism.
Pattern: ['democrats', 'african americans', 'america', 'colorado', 'blacks']↔['systemic racism', 'institutional racism', 'racism']↔['diversity', 'power', 'children', 'inclusion', 'life', 'family', 'world', 'evidence', 'lie', 'fairness']


A radical African American extremist calls for the of white children as the corwd laughs. Racism has no skin color
7:38 PM · May 15, 2023
5.1K Retweets 1.8K Quote Tweets 8.2K Likes


Blacks in Chicago are protesting the illegal invasion of our borders with a fiscally negative impact on black communities. Party of slavery Democrats, who use and abuse the black vote at every election cycle have launched a full on assault against America. Stop voting BLUE
10:50 PM · May 12, 2023
788 Retweets 223 Quote Tweets 2.1K Likes


Remember... "I'm Joe Biden and if you don't vote for me then you ain't black"???? Once a racist always a racist! For those in America who are black...understand one thing...the democrats have hated you since day 1 they are deceitful! #RoseTruths #RoseDC11
2:59 PM · May 1, 2023
1.3K Retweets 82 Quote Tweets 1.3K Likes

BLM.N.1.Race Relations and Systemic Racism in the United States
Narrative summary: America is a racist country with racist leaders, where systemic racism and white nationalism are embedded in society, and those who deny it are complicit in the harm to Black, Latino, Asian, Indigenous, and LGBTQ communities.
Pattern: ['america', 'latin', 'americans', 'biden administration', 'american', 'canada', 'far right', 'united states', 'us leaders']↔['racist', 'discrimination', 'racist country', 'white nationalism', 'systemic racism']↔['nation', 'laws', 'democracy', 'injustice', 'harm', 'justice', 'equality']


In America, white-bodied people enslaved Black-bodied people. Period. Full step. You can't reverse it. Systemic racism flows goes ony way only. So off @RepMTG OFF!!!
6:16 AM · May 19, 2023
2.5K Retweets 780 Quote Tweets 13K Likes


For those who need to hear this: You DO NOT get to tell a Black, Latino, Asian, Indigenous, Jewish or LGBTQ person who tweets something about America's violent racist past that he or she is being divisive and to get over it. These are OUR stories- and they matter! #KeepItWoke
11:23 AM · May 13, 2023
660 Retweets 97 Quote Tweets 2.5K Likes


White nationalism is a political ideology characterized by a belief in the superiority of the white race and a desire to maintain a white-dominated society. Although it has been long present in American society, it has experienced an upsurge in recent years, fueled in part by a growing sense of economic and cultural anxiety pushed by political influence. At its most extreme, white nationalism has manifested in hate crimes and violence directed toward non-white individuals.
11:11 PM · May 19, 2023
1.8K Retweets 166 Quote Tweets 3.5K Likes

BLM.N.2.Criminal Justice and America's Struggle with Crime, Law, and Race
Narrative summary: MAGA Republicans are undermining law enforcement and fostering violence, while systemic racial injustice continues to affect marginalized communities, as seen in the case of George Floyd."
Pattern: ['maga republican', 'maga', 'republicans' 'america']↔['violence', 'murder', 'domestic terrorism', 'crime', 'attack', 'jail', 'governor', 'law']↔['christian', 'accountability', 'safety', 'god', 'justice', 'evidence', 'truth']


Rep. Plaskett: "This hearing is evidence, as if we needed any more, that MAGA Republicans are a threat to the rule of law in America ... my colleagues on the far right are on a mission to attack, discredit, and ultimately dismantle the FBI. This is defund the police on steroids."
2:20 PM · May 18, 2023
1.1K Retweets 107 Quote Tweets 4.4K Likes


MISSING CONTEXT
If we add same-race crime, we see that the vast majority of violent crimes in America involve white victims and white offenders.
If we thought race was a huge factor, and wanted to make the biggest impact on crime numbers, we might want to start there.
5:43 PM · May 8, 2023
1.5K Retweets 294 Quote Tweets 9.8K Likes


40,000 Concentration Camps used to
minorities, Gypsies, Jews, Catholics, Homosexuals, Lesbians, Mentally Ill: we said "NEVER AGAIN" yet America is PROMOTING genicide via MAGA Republican idealogy. It stops. We control the outcome.
7:15 PM · May 14, 2023
731 Retweets 58 Quote Tweets 1.3K Likes

TABLE I
THE MOST VIRAL PAIR OF NARRATIVES AND SAMPLE TRIPLET PATTERNS STRUCTURED AS (NE ↔ IS/AS ↔ VAL) WITH MATCHING SAMPLE TWEETS.

domain experts annotates every phrase within each identified narrative pattern, assigning a stance label ("pro", "anti", or neutral) based on their collective interpretation of the narrative title and summary. Their annotations are then leveraged to generate a weakly supervised training dataset, consisting of 87,359 matches for training the MaskedABSA [45] stance detection model.

We also curate a set of 1,000 random tweets, yielding 2,283 unique masked phrases, to serve as a gold-standard test dataset for labeling and performance assessment of the MaskedABSA stance detector. This manually annotated gold-standard dataset comprises 997 (23%) positive and 3,416 (77%) negative labels.

On this evaluation set, the model achieves an accuracy of 86.68% and an F1 score of 81.10%, compellingly underscoring the effectiveness of our weakly-labeled dataset generation process. A topical breakdown reveals that F1 scores exhibit variability, ranging from a low of 0.57 for the *Crime & Violence* topic (e.g., terms such as "murder" and "police brutality") to a high of 0.89 for the *Religion* topic (e.g., terms such as "Christianity" and "church").

E. Scaling for Users, Messaging and Sub-Communities

To scale users and phrases ideologically, we apply the ANCO-HITS algorithm [43], which extends the classic HITS [41] approach to signed bipartite graphs, allowing simultaneous ranking of users and content based on their mutual stances on a [-1,+1] range such that the position of a user node is closer to the positions of phrases it is connected with positive edges (i.e., indicating support) and further from the positions of phrases it is connected with negative edges (i.e., indicating opposition).

Phrase	**Score ↓**	**Phrase**	**Score ↓**
...		democratic party	0.2401
gun fetish policies	-0.9973	white liberals	0.2771
white media	-0.9972	...	
tommy tuberville	-0.9964	reparations	0.5008
jim jordan	-0.9963	joe biden	0.5581
...		abortion	0.5792
j6ers	-0.7620	miorities	0.6157
republican	-0.7617	...	
tim scott	-0.7614	black vote	0.7109
police officer	-0.7541	black race	0.7129
...		trans	0.7147
george soros	-0.4820	hispanic man	0.7159
mike pence	-0.4722	...	
putin	-0.4340	rep bowman	0.9458
clarence thomas	-0.4245	native americans	0.9462
...		black ranchers	0.9659
illegal aliens	0.2197	martin luther	0.9742
democrat	0.2337	...	

TABLE II
SAMPLE PHRASES RANKED BY THEIR IDEOLOGICAL SCORES ON A $[-1, +1]$ SCALE USING THE ANCO-HITS ALGORITHM. NEGATIVE SCORES INDICATE ALIGNMENT WITH ALM-ASSOCIATED NARRATIVES, WHILE POSITIVE SCORES REFLECT ALIGNMENT WITH BLM-ASSOCIATED NARRATIVES.

We compare the derived rankings against the discerned community/subcommunity structures and their associated camp labels. Validation confirms a robust alignment between the ANCO-HITS scaling and the expert-determined camp codes. Consequently, this approach not only enables the accurate partitioning of users into their respective ideological camps but also affords us the precision to capture fine-grained distinctions in their ideological alignment.

IV. Results & Findings

To assess the efficacy of our scaling methodology, we compare the user rankings generated by the ANCO-HITS algorithm against their assigned subcommunity camp labels. The cutoff used to separate the ANCO-HITS network into two belief-based groups is determined empirically by evaluating different values and choosing the one that maximizes the ratio of appropriate edges to total edges. "Appropriate edges" are counted as the number of green (supporting) edges on one side together with red (opposing) edges on the other.

We rank 115,876 users based on their aggregated stance interactions with 80 actors, 116 issues/aspects, and 53 stance-coded values that emerge from our weakly-labeled data generation process. A cutoff score of 0.29—delineating the boundary between pro-ALM and pro-BLM stance-coded phrases—is applied. This allows for the classification of users: those with scores exceeding this threshold are aligned with the BLM camp, while those falling below it are categorized as aligned with the ALM. Table II presents a sample list of stance-coded phrases ranked by their ANCO-HITS scores.

When compared with labels derived from subcommunity membership, the inferred classifications achieve an accuracy of 90.91%. This indicates a close alignment between our stance-based rankings and the subcommunity labels.

The pattern of the ANCO-HITS score by communities suggests that Community 4, especially subcommunities 4.1 and 4.2, may serve as bridges between ALM- and BLM-aligned groups. Whereas most subcommunities in Communities 2, 3, and 7 cluster on the right of the cutoff and those in Communities 1, 5, and 6 cluster on the left, Community 4 displays a broader, more centered distribution.

Subcommunities 4.1 and 4.2, in particular, have a mean ANCO-HITS score that sits close to the cutoff line, and their standard deviation bars span both sides of the ideological spectrum. This suggests that users within these subcommunities engage across the ideological divide, unlike their other more ideologically siloed counterparts in other communities.

We also leverage θ scores from [1] as an external ground truth for user political alignment assessment. There are 33,371 users overlapping with our dataset where matching users' predicted camp to their θ score sign yields an ALM-camp accuracy of 81.7% and a BLM-camp accuracy of 91.3%. Among misclassified users, within the θ-score interval of $[-0.1, 0.1]$, there are 118 ALM users and 213 BLM users. For θ scores above 0.1 and below -0.1, there are 1,724 ALM users and 1,821 BLM users. We analyze misclassification patterns in both camps and find that one reason is the limited performance of the MaskedABSA model on certain users. Additionally, sarcastic uses of ALM/BLM keywords still trigger the same indicators as genuine endorsements, and without irony detection, the model routinely mistakes sarcasm for support.

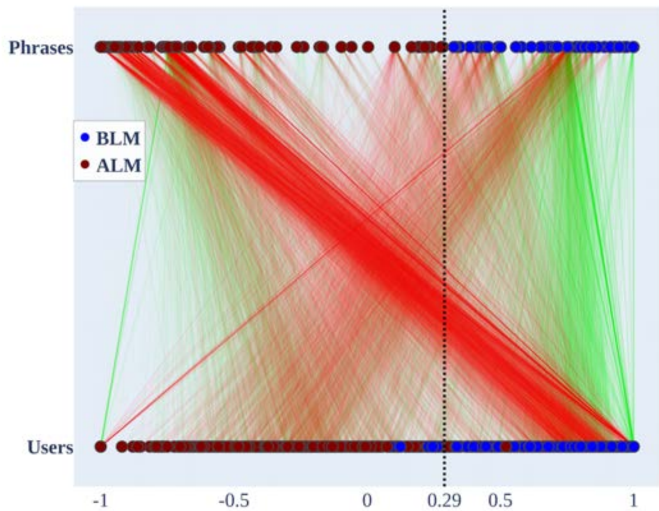

Fig. 2. Sampled users and phrases on ANCO-HITS bipartite graph.

Figure 2 depicts the bipartite graph for 1,000 users uniformly sampled from the dataset. Users labeled as ALM and BLM within the network are represented by red and blue nodes, respectively. Similarly, ALM-aligned phrases and BLM-aligned phrases are represented by red and blue nodes, respectively. Red edges indicate a negative stance, and green edges indicate a positive stance. The visualization corroborates the findings of [56], indicating that Democrats exhibit lower tolerance toward Republicans, as evidenced by the higher density of negative edges from BLM-labeled users towards ALM-labeled phrases.

V. Conclusion

This paper presented a framework for scaling ideological positions in retweet networks by combining community detection, narrative extraction, weakly supervised stance classification, and bipartite ranking. The approach scales both users and phrases, producing interpretable narratives for polarized datasets. It achieves 91% accuracy when validated against subcommunity labels and ideal points from political elites and the broader public across the U.S. and five European nations.

Our results show the benefit of combining stance signals with bipartite scaling methods such as ANCO-HITS. Although MaskedABSA struggles with some narrative themes, particularly sarcasm, the aggregation step in ANCO-HITS helps offset these errors. An analysis of misclassified users indicates that sarcasm directed at opposing narratives is often mistaken for genuine support. A practical extension would be a post-processing step that flags such cases, allowing experts to annotate them as "sarcastic support" or "sarcastic opposition."

We also found that all 52 identified narratives align with an "othering" style of divisive messaging [57]. When shifting focus to the 1,000 most viral messages, only 5% could be considered genuinely cross-partisan. These rare bridging messages typically highlight global rather than U.S.-specific issues, stress universal values like justice, safety, and equality, and use neutral framing that avoids partisan cues.

This work has limitations. We do not examine how user–narrative embeddings evolve over time, suggesting the need for longitudinal studies to assess stability and change in ideological scaling. Broader validation across languages, cultural contexts, polarized communities, and social media platforms is another promising avenue. Beyond political discourse, the framework could also support qualitative studies such as ranking the ideological leanings of news outlets or analyzing interview data. It further offers a way to assess bias in outputs from large language models.

Finally, while these methods provide powerful tools, they should be applied responsibly, with attention to transparency, fairness, and privacy. If used responsibly, ideological scaling can help us better understand polarization without making divisions worse or opening the door to misuse.

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